





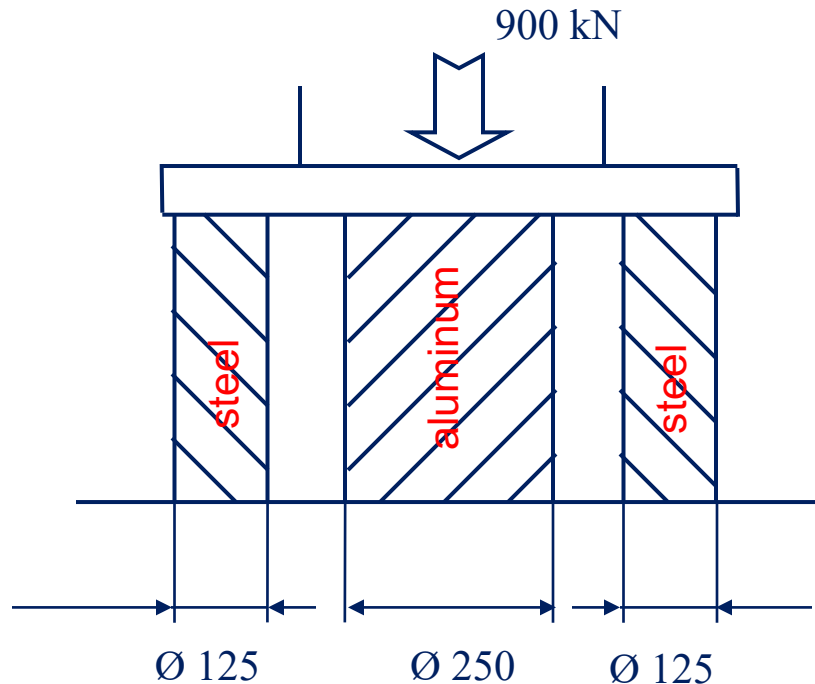
Fundamentals of Metal Forming Numerical Examples



Example 1

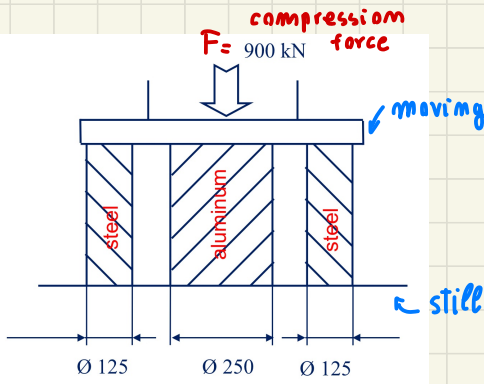
An aluminum cylinder ($D_{\text{aluminum}} = 250 \text{ mm}$) is placed on a non-deformable planar surface with two steel cylinders on the sides ($D_{\text{steel}} = 125 \text{ mm}$).

Determine the (compression stresses)[?] generated in the three parts subject to a load of 900 kN (non-deformable press). Suppose the materials remain in their elastic regions and there is no friction.



$$E_{\text{aluminum}} = 70\,000 \text{ N/mm}^2$$

$$E_{\text{steel}} = 200\,000 \text{ N/mm}^2$$



3 CILINDERS between two dies → uncompressible dies
(very deformable)

$$E_{\text{aluminium}} = 70\,000 \text{ MPa}$$

$$E_{\text{steel}} = 200\,000 \text{ MPa}$$

ELASTIC REGION } hypothesis
NO FRICTION

• compression stresses?

on ELASTIC REGION ⇒ we can refer to $\{S_A, S_S\}$
ENGINEERING / TRUE stress-strain

but $\sigma \approx S$, $\epsilon \approx e$ on elastic

we consider the eng. pnc... on linear region...

S, e → considering engineering values

(b. E)

$$S = E \cdot e \quad \begin{cases} S_A = E_A \cdot e_A \\ S_S = E_S \cdot e_S \end{cases}$$

on el. region, using
HOOKE'S LAW

↑ stress inside the specimen due to the force

knowing that

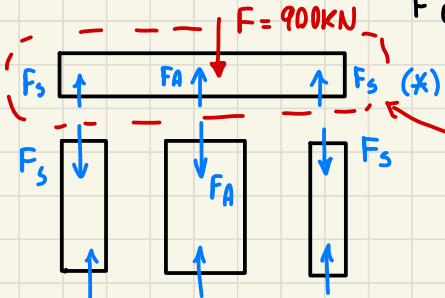
$$S = F/A_0 \quad \begin{cases} S_A = F_A / A_{0,A} \\ S_S = F_S / A_{0,S} \end{cases} \Rightarrow \begin{cases} F_A = S_A A_{0,A} \\ F_S = S_S A_{0,S} \end{cases} (*)$$

↑ FORCE acting on the piece

To find the FORCE...

considering the portion of the die

F divide on all the cylinders



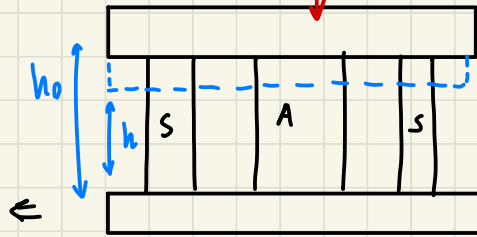
equilibrium on that piece

$$F_A + 2F_S = F \quad (*) \rightarrow \begin{cases} S_A A_{0,A} + S_S A_{0,S} \cdot 2 = F \end{cases}$$

↑ ↑
unknown, we need a second equation!

Situation CONSTRAINTS... $\Rightarrow$ We don't know L_s, L_A cause it is not needed to solve...
all element parallel

$$\begin{cases} S_A A_{0,A} + 2 S_s A_{0,s} = F \\ ? \end{cases}$$



so: they have the SAME STRAIN

$$\parallel \epsilon_A = \epsilon_s = \epsilon \parallel$$

CONSTRAINT ! $\rightarrow$

$$\left\{ \frac{S_A}{E_A} = \frac{S_s}{E_s} \right\} \text{ 2nd equation!}$$

@ end all element has the same h

$$\begin{cases} S_A A_{0,A} + 2 S_s A_{0,s} = F \\ S_s = \frac{E_s}{E_A} S_A \end{cases} \rightarrow \text{from here, 2 eq. with unknown } S_s, S_A$$

$$S_A A_{0,A} + 2 \frac{E_s}{E_A} S_A A_{0,s} = F$$

$$S_A (A_{0,A} + 2 A_{0,s} \frac{E_s}{E_A}) = F \rightarrow$$

$$S_A = \frac{F}{A_{0,A} + 2 A_{0,s} \frac{E_s}{E_A}} = \frac{-900000}{\frac{\pi}{4} 250^2 + 2 \frac{\pi}{4} 125^2 \frac{200K}{70K}} =$$

$$= -7.55 \text{ MPa}$$

$$S_s = \frac{200K}{70K} (-7.55) = -21.57 \text{ MPa}$$

$\downarrow$
hyp. inside
Hooke's Law
is reasonable

NOTE here $\epsilon \approx \epsilon$
Cause on ELASTIC Region, $\Rightarrow$ on PLASTIC region is better to use ϵ TRUE value!!

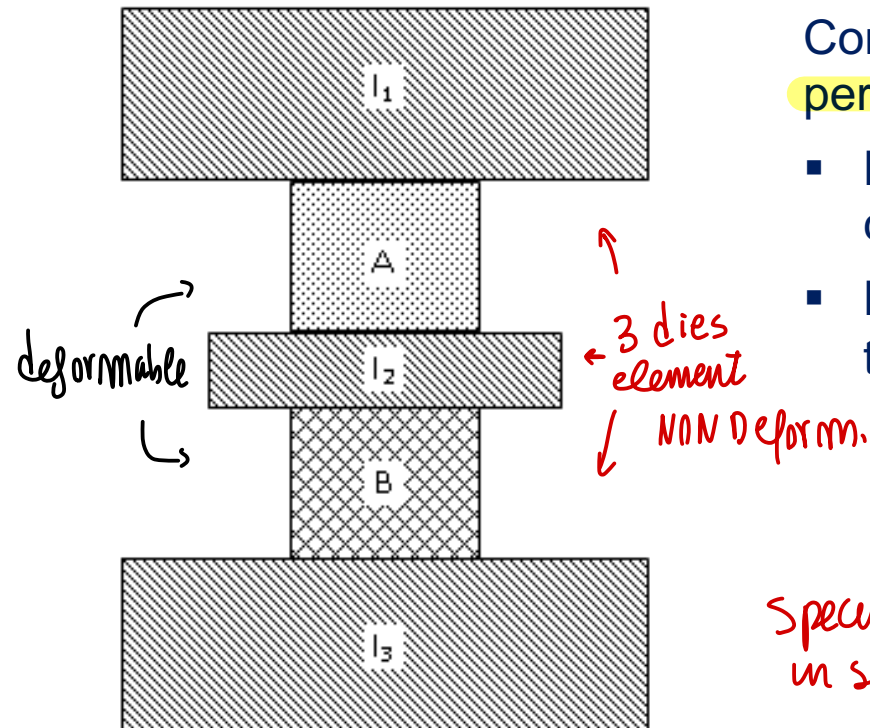


Example 2

3

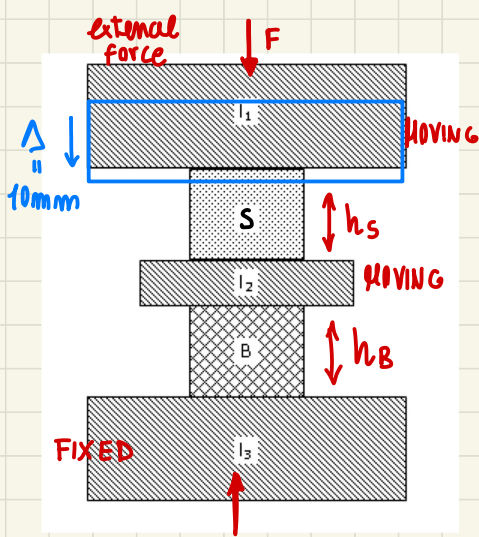
Consider a system consisting of three non-deformable bodies I1, I2, and I3, and two identical cylindrical bodies A and B (height 50 mm and diameter 40 mm), but A is made of steel ($Y = 600$ MPa) and B is made of bronze ($Y = 320$ MPa).

The body I2 is free to move in the vertical direction, the body I3 is fixed, and the body I1 is forced to lower by 10 mm.



Considering the behaviour of A and B as perfectly plastic and neglecting friction:

- Determine the final height of the two cylindrical bodies A and B;
- Determine the force necessary to obtain the 10 mm lowering of the body I1.



S: [steel] $h_0 = 50 \text{ mm}$ $d_0 = 40 \text{ mm}$

$Y_S = 600 \text{ MPa}$

B [broomze] h_0, d_0 same

$Y_B = 320 \text{ MPa}$

{Hyp}

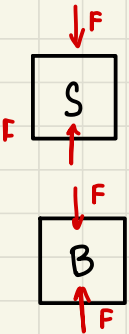
- PLASTIC Region, perfectly PLASTIC (true if they material are HOT $\rightarrow$ recrystallization)
- No Friction

moving I_1 through an external force F (unknown), of $\Delta = 10 \text{ mm}$ displacement

$[h_S, h_B ? F ?] \rightarrow$ final height, force to have that displacement?

What is happening... external forces to a series of material,

if parallel different forces ... if in series different case



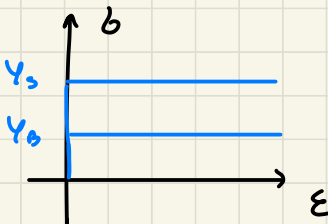
due to equilibrium... same FORCE Acting!

(load to deform material $\Rightarrow$ RIGID dies)

to ANSWER... knowing upper dies moves of $\Delta = 10 \text{ mm}$

we need from h_S, h_B

different properties of S, B $\Rightarrow$ perfectly plastic



$\Leftarrow$ we need to understand the situation...

deformation STARTS for material if we reach Y L_D

SITUATION Q: NO DEF.

SITUATION 1: $F_1 = Y_B A_0$ ←

(If this value reached, we start deformation inside Bronze, lower yield point..)

SITUATION 2: $F_2 = Y_S A_0$

(force that enable to deform steel..)

from this three situation...

IF: ~~$F < F_1$ NO DEF.~~ NOT POSSIBLE
IF: $F_1 \leq F < F_2$ DEF. only B ← we are here
IF: $F \geq F_2$ DEF of B & S

To start deformation, because

$$\sigma = F/A \quad F = \sigma A \rightarrow$$

start Bronze def, $F_1 \geq Y_B A_{0B}$
 $\{A_{0S} = A_{0B} = A_0\}$ F_0 to deform

IF NO DEF., $\Delta = 0$ against the data of the problem, we have movement!

$$A_0 = A_{0B} = A_{0S} = \frac{\pi}{4} 40^2 = 1257 \text{ mm}^2$$

$$F_1 = Y_B A_0 = 402\,240 \text{ N}$$

Bronze

$$F_2 = Y_S A_0 = 754\,200 \text{ N}$$

STEEL

in which situation we are?

evaluating the deformation, STRAIN variation (h variation) in case before we reach F_2 full B deformation and S NOT started

$$\sigma_B = \frac{F_B}{A_B} = \frac{F_2}{A_B} \leftarrow \text{apply } F_2, \text{ because PERFECT PLASTIC}$$

$$\sigma_B = Y_B \rightarrow \frac{F_2}{A_B} = Y_B$$

$$A_B = \frac{F_2}{Y_B} = 2357 \text{ mm}^2 \text{ (actual area)}$$

because VOLUME is constant

$$A_0 h_0 = A_B h_B \rightarrow h_B = \frac{A_0}{A_B} h_0 = 26.67 \text{ mm}$$

$$\Delta = h_B - h_0 = -23.33 \text{ mm}$$

$|\Delta| > 10 \text{ mm}$
so $F < F_2$

→ If we have $F > F_2$ too much Δ ...

So we are on SIT. 1

→ only BRONZE is deforming

and knowing $\Delta = 10 \text{ mm}$, we can compute again...

$$\Delta h = -10 \text{ mm} = h_B - h_0 \rightarrow h_B = h_0 - 10 = 50 - 10 = 40 \text{ mm}$$

$$\text{While S is frozen, no DEF.} \rightarrow h_S = 50 \text{ mm}$$

to evaluate F needed

$$F = Y_B A_B = 502800 \text{ MN} = 502.8 \text{ kN}$$

$$\bar{V} = \text{const} : A_B h_B = A_0 h_0 \rightarrow A_B = A_0 \frac{h_0}{h_B} = 1571.25 \text{ mm}^2$$



Example 3

A cylindrical sample in 1100-O aluminum alloy ($k = 180$ MPa, $n = 0,20$) with an initial height $h_0 = 100$ mm and an initial diameter $d_0 = 80$ mm, is compressed to a final height equal to $h_2 = 40$ mm in two steps between two flat dies with zero friction.

Determine:

- the engineering and true strains at the end of each step ($h_1 = 70$ mm, $h_2 = 40$ mm);
- the force required for each step;
- the strain energy density and the ideal work spent on the overall deformation.

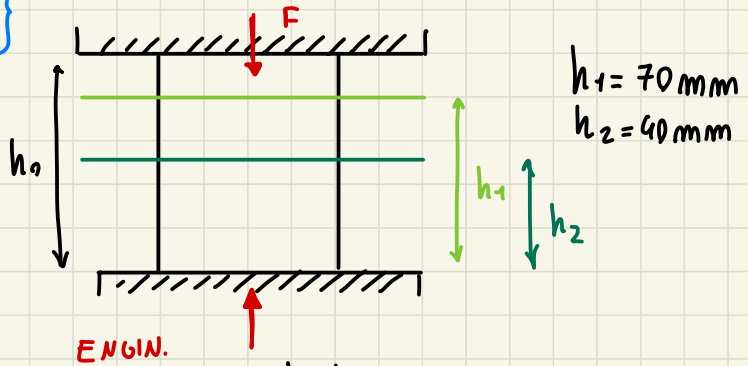
ci-inducal sample, ALUMINIUM Alloy

FLOW CURVE
 $\sigma = K \epsilon^m$

$K = 180 \text{ MPa}$
 $m = 0.2$
 $d_0 = 80 \text{ mm}$
 $h_0 = 100 \text{ mm}$

open die forging
 in 2 steps...
 $h_0 \rightarrow h_1 \rightarrow h_2$

$\epsilon, \epsilon ? F ? \mu, W_i ?$
 for each step for overall deformation



ENGINE

$$\epsilon_{0 \rightarrow 1} = \frac{h_1 - h_0}{h_0} = -0.3 = -30\%$$

$$\epsilon_{1 \rightarrow 2} = \frac{h_2 - h_1}{h_1} = -0.4286 = -42.86\%$$

$$\epsilon_{0 \rightarrow 2} = \frac{h_2 - h_0}{h_0} = -0.6 = -60\%$$

each step

overall

as you can see
 $\epsilon_{0 \rightarrow 1} + \epsilon_{1 \rightarrow 2} \neq \epsilon_{0 \rightarrow 2} !$

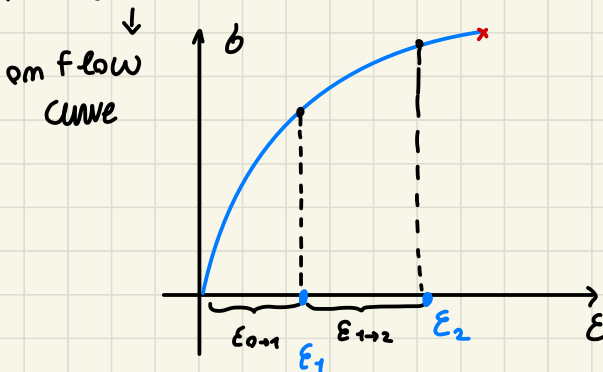
TRUE STRAIN

$$\epsilon_1 = \epsilon_{0 \rightarrow 1} = \ln \left(\frac{h_1}{h_0} \right) = -0.3567 = -35.67\% \neq \epsilon_{0 \rightarrow 1}$$

$$\epsilon_{1 \rightarrow 2} = \ln \left(\frac{h_2}{h_1} \right) = -0.5596 = -55.96\%$$

$$\epsilon_2 = \epsilon_{0 \rightarrow 2} = \ln \left(\frac{h_2}{h_0} \right) = -0.9163 = -91.63\% = \epsilon_{0 \rightarrow 1} + \epsilon_{1 \rightarrow 2}$$

speaking True strain...



summy true strain get the total !

for different step def. the true strain is the summy single steps strain

Force required $\forall$ step $\rightarrow$ We need the AREA...



$$\sigma = F/A \rightarrow F = \sigma \cdot A \quad (\forall = \text{constant})$$

step 1) $h_0 A_0 = A_1 h_1 \rightarrow A_1 = A_0 \frac{h_0}{h_1} = 7181 \text{ mm}^2$

$$A_0 = \frac{\pi}{4} 80^2 = 5027 \text{ mm}^2$$

step 2) $h_0 A_0 = A_2 h_2$

$$A_2 = A_0 \frac{h_0}{h_2} = 12568 \text{ mm}^2$$

from the strain values...

$$\left\{ \begin{array}{l} \epsilon_1 = -0.3567 \\ \epsilon_{1 \rightarrow 2} = -0.5596 \\ \epsilon_2 = -0.9163 \end{array} \right\}$$

and from Flow CURVE expression...

$$\sigma_1 = K \epsilon_1^m = -146.46 \text{ MPa}$$

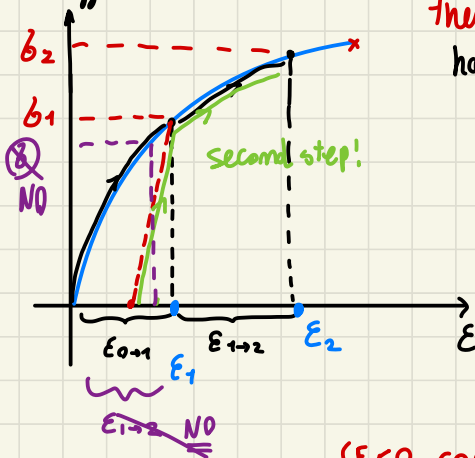
$$\sigma_2 = K (\epsilon_2)^m =$$

Carefull, when evaluating σ_2 you may use all ϵ_2

if two steps in different machine $\rightarrow$ first σ_1 , then elastic recovery

then deformed until σ_2 , material has new behavior, NO use $\epsilon_{1 \rightarrow 2}$ to the σ_2 !

You need to deform up to ϵ_2



$$\sigma_2 = K \epsilon_2^m = -176.88 \text{ MPa}$$

increasing deformation! we need more stress!

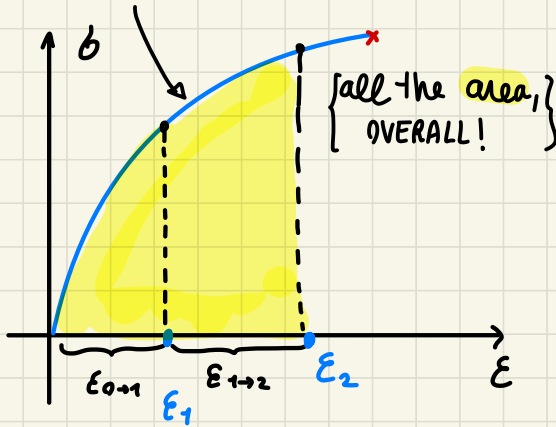
($F < 0$, compression)

$$\Rightarrow F_1 = \sigma_1 A_1 = -1051729,26 \text{ N} = -1.052 \text{ MN}$$

$$F_2 = \sigma_2 A_2 = -2223027,84 \text{ N} = -2,223 \text{ MN}$$

Force in hand to reach ϵ_2

overall strain energy density, ideal work



losses neglected

$$u = \int_0^{\epsilon_2} \sigma d\epsilon = \int_0^{\epsilon_2} K \epsilon^m d\epsilon =$$

$$= \frac{K \epsilon_2^{m+1}}{m+1} = 135 \text{ MPa}$$

to transform as energy J/mm³

$$u = 135 \text{ MPa} = \frac{135}{1000} \frac{\text{J}}{\text{mm}^3} =$$

$$= 0.135 \text{ J/mm}^3$$

$\bar{V}$ constant

$$W_i = u \bar{V} = 0.135 \cdot (A_0 \cdot h_0) = 67865 \text{ J} = 67.9 \text{ kJ}$$

MORE STEPS! $\Rightarrow$ material changes behaviour

you need to consider the total ϵ_2 , refer to
add each deformation step, all CURVE VSTEP



Example 4

Metal Forming

↑ NOT to make Holes...
from cylinder

HOT WORKING
open die forging

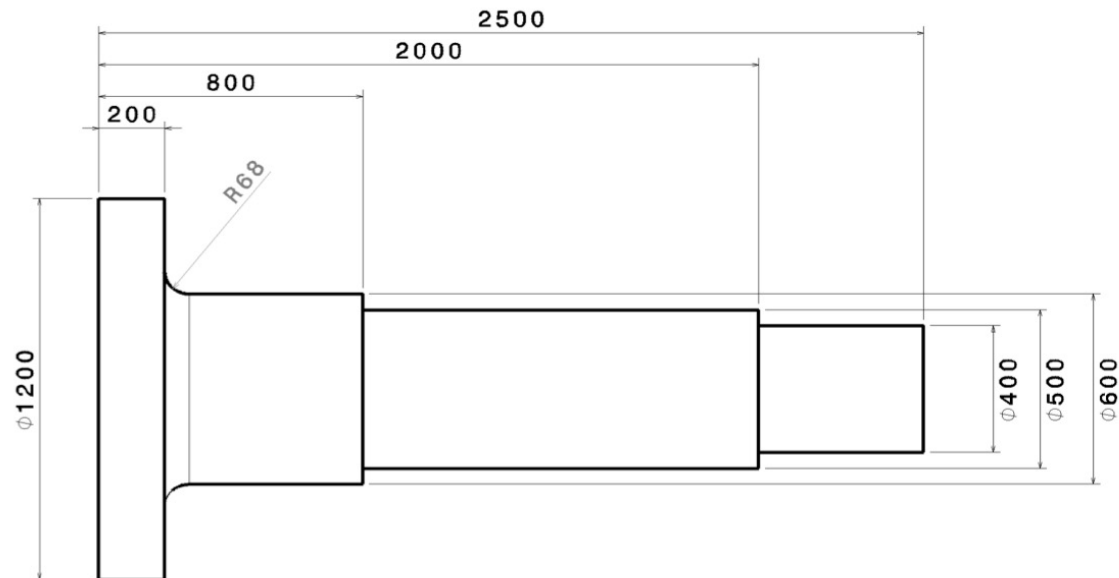
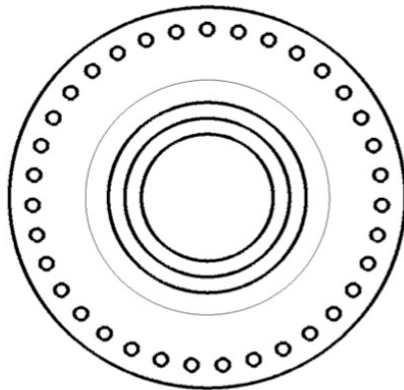
5

compression methods

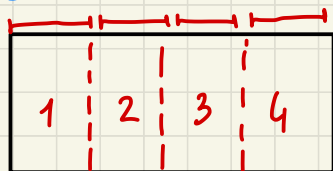
The production process of the shaft in the figure involves a set of forging operations and a subsequent upsetting of the flange. The process takes place at a temperature such as to guarantee a perfectly plastic behaviour with $Y = 180$ MPa.

Assuming starting from a cylinder with a diameter of $d_0 = 800$ mm, determine the ideal work required to obtain each element of the shaft. → each shape

real
component



starting point

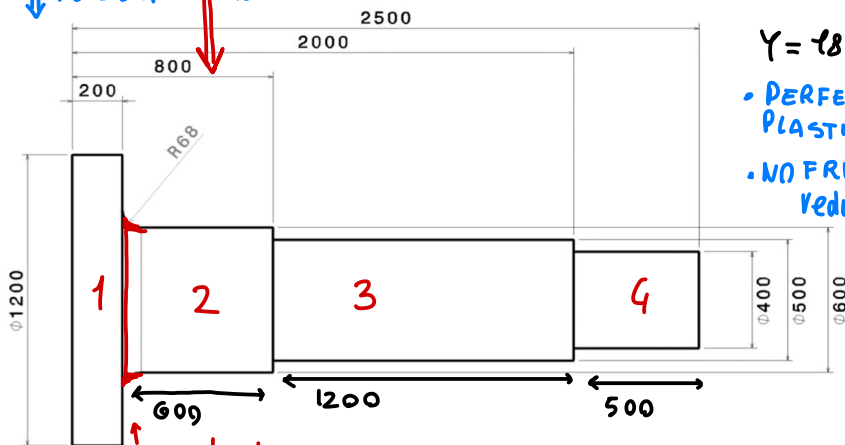


$\phi D_0 = 800 \text{ mm}$

ideal work? to shape from the initial object
↓
w V operation

deformation from here

↓ to OBTAIN that SHAFT



$\gamma = 180 \text{ MPa}$

- PERFECTLY PLASTIC
- NO FRICTION, redundant work

considered as a cylinder → too complex shape



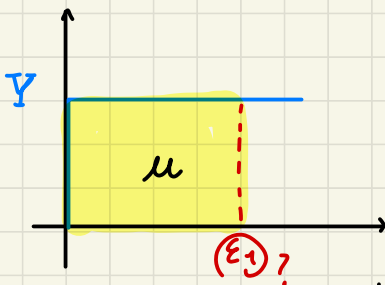
↓
We need to understand

define the ideal work

V OPERATION → (V element)

$$W_i = \mu V$$

↓ STRAIN energy density



ideal situation
PERFECTLY PLASTIC

E strain to shape each element

neglect the sign (absolute value)
↓

$$\mu = |\gamma| |\epsilon_1|$$

UNKNOWN

$$|\epsilon| = \left| \ln \left(\frac{l_1}{l_0} \right) \right|$$

final length

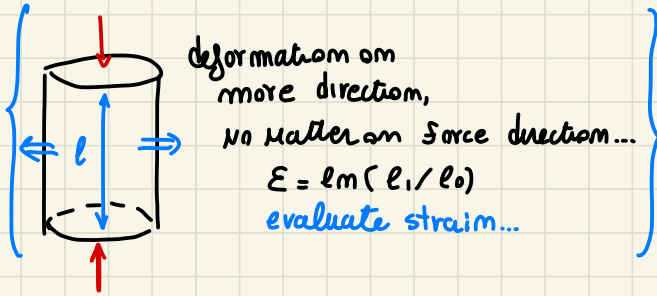
starting point of cylinder unknown

← UNKNOWN?

we divide the cylinder, we know $l_{1,2,3,4}$ final but NOT starting l_0 V element

once evaluated ϵ (known l_0 $\forall$ element) $\rightarrow$ so u_i $\forall i=1,2,3,4$

$\downarrow$ and from here
 $\forall i=1,2,3,4$ to evaluate w_i



to evaluate l_0 of each part, I may consider the constant volume
 $V = \text{constant}$

same logic $\forall$ part

$$1) \frac{\pi}{4} d_{0,1}^2 l_{0,1} = \frac{\pi}{4} d_{1,1}^2 l_{1,1} \rightarrow l_{0,1} = l_{1,1} \frac{d_{1,1}^2}{d_{0,1}^2} = 200 \frac{1200^2}{800^2} = 450 \text{ mm}$$

(1) is COMPRESSED to increase the diameter

$$2) l_{0,2} = l_{1,2} \frac{d_{1,2}^2}{d_{0,2}^2} = 600 \frac{600^2}{800^2} = 337.5 \text{ mm}$$

... @ same way

all same
 $d_{0,i}$ CYLINDER

$$3) l_{0,3} = 1200 \frac{500^2}{800^2} = 468.75 \text{ mm}$$

$\Rightarrow$ from the sum of
 all element

$$4) l_{0,4} = 500 \frac{400^2}{800^2} = 125 \text{ mm}$$

$$L_0 = 1381.25 \text{ mm}$$

total CYLINDER lenght

different deformation

$\forall$ pieces to obtain all
 the parts

Known $l_0 \rightarrow$ We can compute
 all the STRAIN... of each
 element $j=1,2,3,4$ ϵ_j

$$\epsilon_1 = \left| \ln \left(\frac{200}{450} \right) \right| = 0.8109$$

$$\epsilon_3 = \left| \ln \left(\frac{1200}{468.75} \right) \right| = 0.94$$

$$\epsilon_2 = \left| \ln \left(\frac{600}{337.5} \right) \right| = 0.5754$$

$$\epsilon_4 = \left| \ln \left(\frac{500}{125} \right) \right| = 1.3863$$

strain energy density $\forall$ piece

$$\mu_1 = |\gamma| |\epsilon_1| \cdot 1/1000 = 0.146 \text{ J/mm}^3$$

$$\mu_2 = |\gamma| |\epsilon_2| \cdot 1/1000 = 0.104 \text{ J/mm}^3$$

$$\mu_3 = |\gamma| |\epsilon_3| \cdot 1/1000 = 0.169 \text{ J/mm}^3$$

$$\mu_4 = |\gamma| |\epsilon_4| \cdot 1/1000 = 0.25 \text{ J/mm}^3$$

to change from
MPa to J/mm³

Volume $\forall$ piece

$$V_1 = \frac{\pi}{4} 1200^2 200 = 226194671 \text{ mm}^3$$

$$V_2 = \frac{\pi}{4} 600^2 600 = 169646003 \text{ mm}^3$$

$$V_3 = \frac{\pi}{4} 500^2 1200 = 235619449 \text{ mm}^3$$

$$V_4 = \frac{\pi}{4} 400^2 500 = 62831853 \text{ mm}^3$$



so knowing μ_i , V_i we can compute the ideal work

$$W_{i1} = \mu_1 V_1 \approx 33 \text{ MJ}$$

$$W_{i2} = \mu_2 V_2 \approx 17.6 \text{ MJ}$$

$$W_{i3} = \mu_3 V_3 \approx 39.8 \text{ MJ}$$

$$W_{i4} = \mu_4 V_4 \approx 15.7 \text{ MJ}$$

ideal work to shape
the cylinder
by HOT WORKING
process!

↓ by summing W_{i1} we obtain
all the work $W_{\text{tot}} = \sum_i W_{i1}$

$\epsilon \approx \epsilon$ on ELASTIC Region

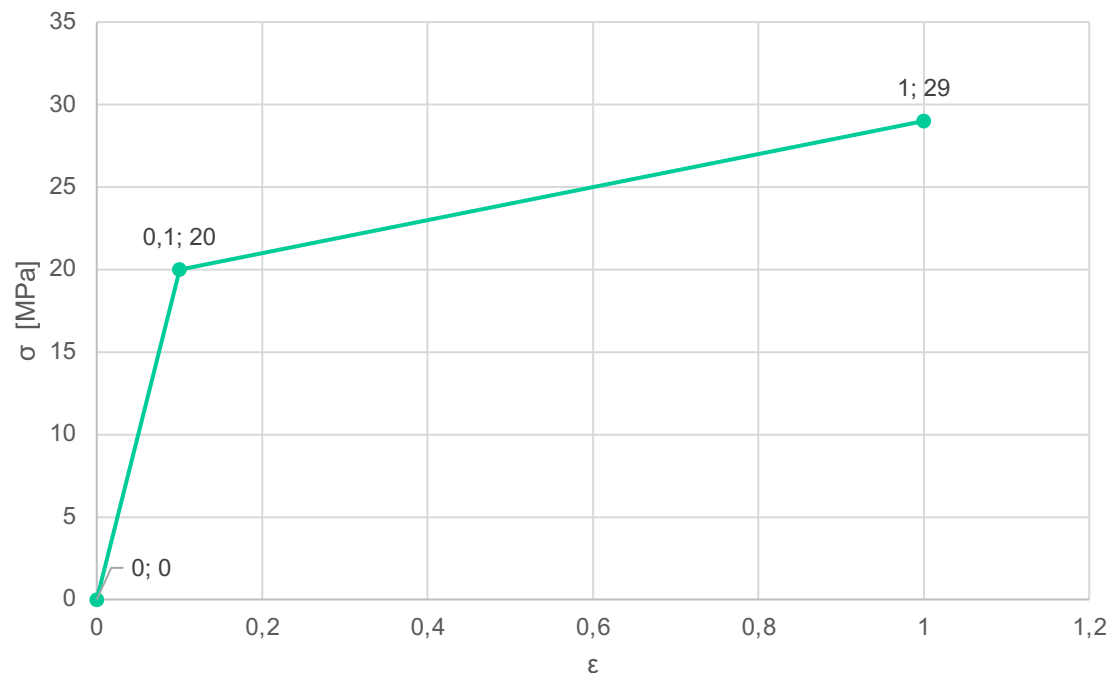
$\epsilon \napprox \epsilon$ on PLASTIC Region ϵ don't Approx Well $\rightarrow$ Use true stress/STRAIN
on PLASTIC def. !



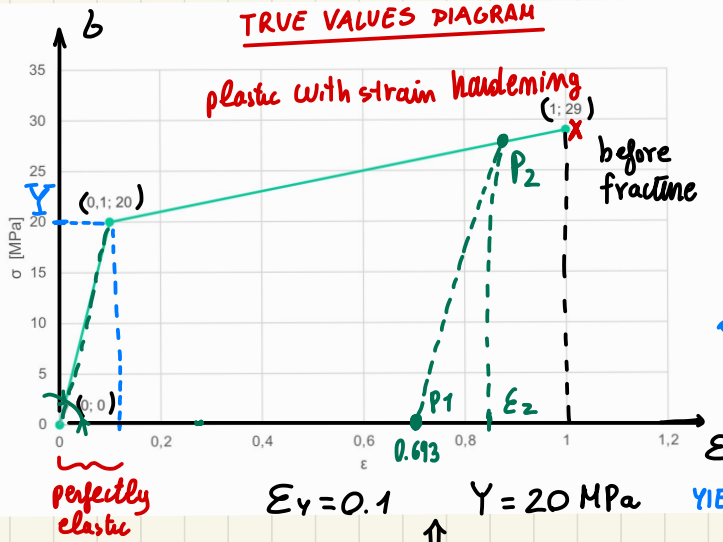
Example 5

A cylindrical specimen, with an initial height of 40 mm, is compressed with the aim of reaching a final height of 20 mm.

Considering the elastic recovery and the behaviour of the material as shown in the figure, what should be the actual height at which the specimen must be compressed?



TRUE VALUES DIAGRAM



cylindrical specimen
 $h_0 = 40 \text{ mm}$
 compressed with
 All of reaching after
 Metal work
 $h_1 = 20 \text{ mm}$
 (50% compressed)

← MATERIAL
 behav.
 (elastic recovery)

considering elastic recovery + this behav.

actual h to obtain h_1 ?
 reached by specimen
 compressed

↓
 final after
 recovery!

firstly, remaining on the curve...

whatever is the slope → we know that
 after recovery P_1 is characterized by

$$P_1(\epsilon_1, 0) \rightarrow \epsilon_1 = \left| \ln \left(\frac{h_1}{h_0} \right) \right| = \left| \ln \left(\frac{20}{40} \right) \right| = 0.693$$

deform, considering
 that UNLOADING,
 the behav. $\approx$ HOOKE LAW when recovery

We need to reach point (P_2) , so ϵ_2

h : final height → evaluating ϵ_2 !

(simply by intersecting the line...)

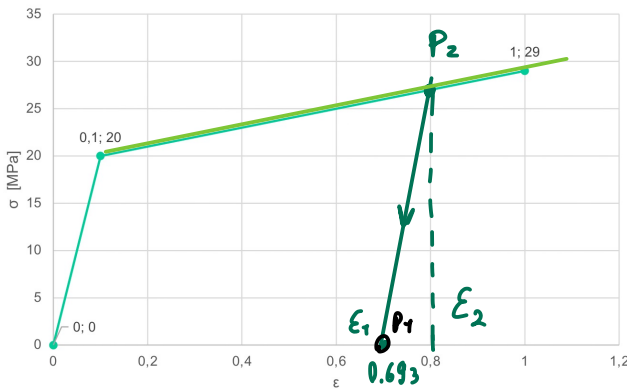
slope E :
 ⇒ from elastic
 region

$$E = \frac{Y}{\epsilon_1} = \frac{20}{0.1} = 200 \text{ MPa}$$

slope of
the curve

equation of STRAIN HARDENING line

and RECOVERY line → intersection to find P_2 !



Recovery line

$$\sigma - \sigma_1 = E(\epsilon - \epsilon_1)$$

$$\sigma = 200(\epsilon - 0.693)$$

$$\boxed{\sigma = 200\epsilon - 138.6}$$

strain hardening line

$$\sigma - \sigma_1 = \frac{\sigma_3 - \sigma_1}{\epsilon_3 - \epsilon_1}(\epsilon - \epsilon_1)$$

$$\sigma = 20 + \frac{29 - 20}{1 - 0.1}(\epsilon - 0.1)$$

$$\boxed{\sigma = 10\epsilon + 19}$$

by intercepting this two lines

We obtain P_2



$$\begin{cases} \sigma = 200\epsilon - 138.6 \\ \sigma = 10\epsilon + 19 \end{cases}$$



$$200\epsilon - 138.6 = 10\epsilon + 19 \rightarrow \epsilon_2 = \frac{157.6}{190} = 0.8295$$

so from the Real strain

value ϵ_2 we can obtain h_2 , height to which compress

$$\epsilon_2 = \ln(h_2/h_0)$$



$$h_2 = h_0 e^{\epsilon_2} = 40 e^{-0.8295} = 17.45 \text{ mm}$$

compression!
so it is a REDUCTION of h

↑
required final height
before elastic recovery



Example 6

The height of an annealed copper cylinder with a diameter $d_0 = 25$ mm is reduced by 70% starting from a value $h_0 = 50$ mm by means of a metal forming process.

The material has a flow curve with parameters $k = 315$ MPa and $n = 0,54$.

Neglecting the barrelling phenomenon, determine / estimate:

- The maximum force in the absence of friction;
- The maximum force with friction equal to $\mu = 0,25$;
- The maximum force with friction equal to $\mu=0,5$;
- The strain energy density and the ideal work required to complete the forging.

Cylinder

$$d_0 = 25 \text{ mm}$$

$$h_0 = 50 \text{ mm}$$

- 70% (Reduction)

Flow Curve
behav. $\begin{cases} K = 315 \text{ MPa} \\ m = 0.54 \end{cases}$

$$\sigma = K \epsilon^m$$

Borelling Neglected (redundant def)

$$F? \rightarrow \begin{cases} \mu = 0 \\ \mu = 0.25 \\ \mu = 0.5 \end{cases}$$

$\mu, W_i?$ to perform the operation

FORCE: to evaluate the force

$$F = p_{av} A_1 \quad \text{@ end of deform.}$$

on that situation $p_{av} = b_1 \left(1 + \frac{\mu d_1}{3 h_1} \right)$

final values..

$$h_1 = (1 - 0.7) h_0 = 0.3 \cdot 50 = 15 \text{ mm}$$

$d_1 = ?$ Geometry does NOT change...
same final shape

$$\frac{\pi}{4} d_0^2 h_0 = \frac{\pi}{4} d_1^2 h_1 \quad V = \text{constant}$$

$$A_1 = \frac{\pi}{4} (45.644)^2 = 1636.28 \text{ mm}^2$$

$$d_1 = d_0 \sqrt{\frac{h_0}{h_1}} = 25 \sqrt{\frac{50}{15}} = 45.644 \text{ mm}$$

$$\sigma_1 = K \epsilon_1^m$$

$$\epsilon_1 = \ln \left(\frac{h_1}{h_0} \right) = -1.2$$

$$\sigma_1 = 315 (-1.2)^{0.54} = -347.59 \text{ MPa}$$

↓
So we have all the data to evaluate the force $\forall$ combination of μ

$\mu = 0$:

$$p_{av} = \sigma_1 = 347.59 \text{ MPa} \rightarrow F = p_{av} A_1 = 568.8 \text{ kN}$$

ideal combination
($\mu = 0$)

force to have full deformation with UNFRICTION

$\mu = 0.25$:

$$p_{av} = \sigma_1 \left(1 + \frac{\mu d_1}{3 h_1} \right) = 435.73 \text{ MPa} \rightarrow F = p_{av} \frac{A_1}{1000} = 713 \text{ kN}$$

$\mu = 0.5$

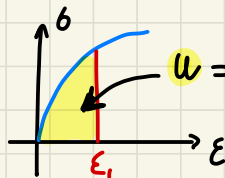
(same way...)

$$p_{av} = 523.87 \text{ MPa}$$

$$\rightarrow F = 857.2 \text{ kN}$$

$$V_0 = \frac{\pi}{4} (25)^2 50 = 24544 \text{ mm}^3$$

STRAIN energy density $\Rightarrow$ as integral of flow curve



$$u = \frac{K \epsilon^{m+1}}{m+1} \quad \frac{1}{1000} = 0.271 \frac{\text{J}}{\text{mm}^3}$$

to have J/mm³

$$W = u V_0 = 6651 \text{ J}$$



Example 7

A **cylindrical rivet** in bronze C74500, with an initial diameter $d_0 = 3$ mm, is compressed to a final diameter equal to $d_1 = 6$ mm and a final height equal to $h_1 = 3$ mm.

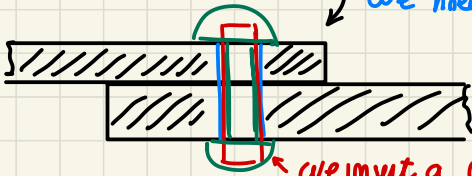
The process takes place at a temperature such as to guarantee a perfectly plastic behaviour with $Y = 170$ MPa.

Assuming a friction coefficient $\mu = 0,2$, determine the required force.

RIVET is a Joim process (HOT WORKING)

two elements to join...

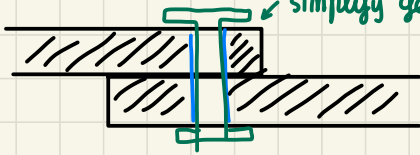
we holes the pieces to join



we insert a cylindrical element inside

then we deform the two sides to join the parts!

plastic deformation process on the cylinder outside → connecting the parts



simplify geometry

HOT WORKING,

|| perfectly plastic behav. ||

on the initial cylindrical piece to be deformed

$$d_0 = 3 \text{ mm} \rightarrow d_1 = 6 \text{ mm}$$

$$h_1 = 3 \text{ mm}$$

looking on one side of RIVET

$$Y = 170 \text{ MPa} \quad \mu = 0.2 \quad F?$$

PERFECTLY
PLASTIC

$$P_{AV} = Y \left(1 + \frac{\mu d_1}{3 h_1} \right) = 170 \left(1 + \frac{0.2 \cdot 6}{3 \cdot 3} \right) = 192.67 \text{ MPa}$$

we need ↑
to reach Y to deform!

$$A_1 = \frac{\pi}{4} 6^2 = 28.3 \text{ mm}^2$$



$$\text{so } F = P_{AV} \cdot A_1 = 5453 \text{ N}$$



Example 8

The height of an annealed carbon steel cylinder with a diameter $d_0 = 50$ mm is reduced by 20% starting from a value $h_0 = 100$ mm by open-die forging.

The material has a flow curve with parameters $k = 530$ MPa and $n = 0,26$, and UTS equal to 373 MPa.

Neglecting the barrelling phenomenon, determine / estimate:

- The maximum force with either no friction or friction equal to $\mu = 0,2$;
- The strain energy density and the ideal work required to complete the forging;
- The minimum possible final height. → JUST SOLVING THIS...

cylinder

Neglecting Barrelling /
Redundant work

$$d_0 = 50 \text{ mm}$$

$$h_0 = 100 \text{ mm}$$

-20% (Reduction)

$$\text{FLOW CURVE} \quad \left\{ \begin{array}{l} K = 530 \text{ MPa} \\ b = K \epsilon^m \end{array} \right. \quad m = 0.26$$

Known

$$UTS = 363 \text{ MPa}$$

$$1) F \rightarrow \begin{array}{l} \mu = 0 \\ \mu = 0.2 \end{array} ?$$

$$2) \quad u, W_i ?$$

$$\text{Solve this} \rightarrow 3) h_{\min} \leftarrow \text{possible min } h \text{ that we can reach}$$

We have to
evaluate the

MAX STRAIN possible as plastic
deformation... $|\epsilon_{\max}|$

max
compression
possible!

$$\text{Once evaluated } \epsilon_{\max} \rightarrow h_{\min} = h_0 e^{\epsilon_{\max}}$$

→ from UTS... evaluate recovery → strain related to UTS

will be the maximum ϵ possible

Referring to flow curve

$$UTS = K \epsilon_{\max}^m \rightarrow \epsilon_{\max} = \left[\frac{UTS}{K} \right]^{1/m} = 0.259$$

$$\text{so } h_{\min} = 100 e^{-0.259} = 77.18 \text{ mm}$$

here:

$$h_1 = (1 - 0.2) 100 = 80 \text{ mm} \rightarrow \text{on plastic region}$$

other questions

as other problems

We don't reach an h related
to UTS